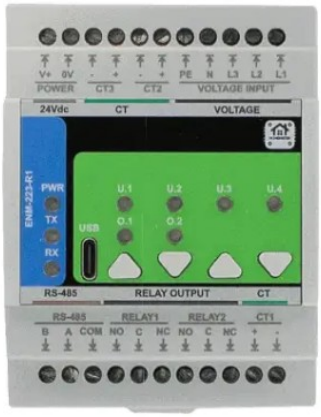
Datasheet

ENM-223-R1 3-phase Energy Metering Module



Part No.: ENM-223-R1

Manufacturer: ISYSTEMS AUTOMATION S.R.L.

Hardware Version: V1.0

Architecture: Modular (MCU Board +Relay board+ Field Board)

Controller: RP2350A

Functional Overview

The ENM-223-R1 is a modular 3-phase energy metering + I/O module for power monitoring and automation. It measures 3 voltage channels (L1/L2/L3-N) and 3 current channels (external CTs) using a dedicated ATM90E32AS metering IC, and provides 2 SPDT relay outputs, plus optional 4 buttons / 4 user LEDs for local control and indication.

The system is implemented using three hardware boards:

Board	Function
MCU Board	CPU, RS-485, USB-C, power conversion (24 V → 5 V → 3.3 V)
Field Board	3-phase metering front-end (voltage dividers + CT inputs), isolated analog domain, field connectors
Relay Extension	2 × SPDT relay output stage

The module connects to a PLC or SCADA as a Modbus RTU slave over RS-485 and is configured via USB-C Web Serial (WebConfig).

Technical Specifications

Specification	Details
Microcontroller	RP2350A dual-core microcontroller
Storage	External QSPI Flash (W25Q32JV)
Power Input	24 V DC nominal
File system	LittleFS persistent configuration storage
Document revision	DS-ENM-223-R1 Rev. B · 2026-09 · Hardware R1 (V1.0)
Metering IC	ATM90E32AS (3-phase energy metering / power quality)
Voltage Inputs	3 × L1/L2/L3-N+PE sensing, 85–265 V AC (mains measurement inputs)
Current Inputs	3 × CT channels (external current-output CTs); secondary ≤ 60 mA RMS into a 6 Ω on-board burden. Full scale 120 / 60 / 30 mA at PGA ×1 / ×2 / ×4 (default ×2). Do not use 1 A or 5 A secondary CTs.
Relay outputs	2 × SPDT (NO / C / NC); 3 A @ 250 VAC per contact (module limit)
Isolation	5 kV RMS between MCU and measurement side (ISO7761); RS-485 not isolated
User Interface	4 buttons, 9 LEDs (power, 4 user, RX, TX, 2× relay)
RS-485	half-duplex Modbus RTU, not galvanically isolated
USB	USB-C (ESD protected, configuration and firmware upload)
Modbus defaults	Address 3, 19200 baud, 8N1

Top Terminal Block (1–11)

Pos	Marking	Signal	Description
1	V+	POWER	24 V DC input
2	0V	POWER	24 V DC return
3	-	CT3	CT3 negative
4	+	CT3	CT3 positive
5	-	CT2	CT2 negative
6	+	CT2	CT2 positive
7	PE	VOLTAGE_INPUT	Protective earth (MAINS)

8	N	VOLTAGE_INPUT	Neutral (MAINS)
9	L3	VOLTAGE_INPUT	Phase 3 sensing (MAINS)
10	L2	VOLTAGE_INPUT	Phase 2 sensing (MAINS)
11	L1	VOLTAGE_INPUT	Phase 1 sensing (MAINS)

Bottom Terminal Block (1–11)

Pos	Marking	Signal	Description
1	B	RS485	RS-485 data -
2	A	RS485	RS-485 data +
3	COM	RS485	RS-485 signal reference
4	NO	RELAY1	Relay 1 normally open
5	C	RELAY1	Relay 1 common
6	NC	RELAY1	Relay 1 normally closed
7	NO	RELAY2	Relay 2 normally open
8	C	RELAY2	Relay 2 common
9	NC	RELAY2	Relay 2 normally closed
10	+	CT1	CT1 positive
11	-	CT1	CT1 negative

*Relay outputs are **not internally fused**. External overcurrent protection must be provided in accordance with applicable standards and installation requirements. Use appropriately rated fuses or circuit breakers and consider external contactors for loads exceeding 3 A or for inductive and high inrush loads.

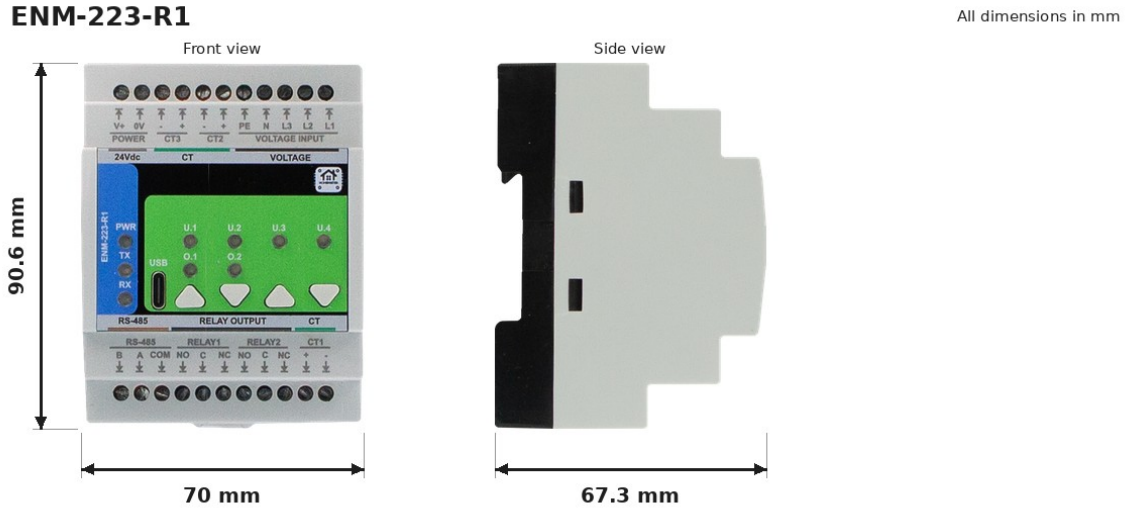
Installation, Environmental & Mechanical

Category	Specification	Details
Environmental Ratings	Operating temperature	0 °C to +40 °C

Environmental Ratings	Storage temperature	−10 °C to +55 °C
Environmental Ratings	Relative humidity	0–90 % RH, non-condensing
Environmental Ratings	Ingress protection	IP20 (inside cabinet only)
Environmental Ratings	Installation	Indoor control cabinet only; not for outdoor or exposed installation
Environmental Ratings	Maximum altitude	2000 m
Environmental Ratings	Pollution degree	2
Mechanical & Packaging	Dimensions	70 × 90.6 × 67.3 mm (L × W × H)
Mechanical & Packaging	DIN width	4 modules (≈ 70 mm)
Mechanical & Packaging	Mounting	35 mm DIN rail
Mechanical & Packaging	Enclosure	PC/ABS industrial enclosure
Mechanical & Packaging	Terminal type	Pluggable screw terminal blocks, 5.08 mm pitch
Mechanical & Packaging	Wire cross-section	0.2–2.5 mm² (AWG 24–12)
Mechanical & Packaging	Tightening torque	0.4–0.6 Nm
Mechanical & Packaging	Net weight	200 g
Mechanical & Packaging	Gross weight	320 g
Mechanical & Packaging	Pack size	140 × 125 × 94 mm (L × W × H)

Install only inside a control cabinet with ventilation; the cabinet must include a protective front plate covering all module connection terminals and a closing protective door; not for outdoor or exposed installation.

All wiring terminals must be protected against accidental contact by an insulating front plate, wiring duct, or terminal cover. Exposed live terminals are not permitted.



Cable Recommendations & Shield Grounding

Use shielded, twisted constructions and bond shields correctly to reduce EMI and ground-loop issues.

General Routing Rules

- Route low-level signal cables separately from mains, relay outputs, contactors, VFD motor cables, and power wiring.
- If crossing power cables is unavoidable, cross at **90°**.
- Keep cable runs as short as practical and avoid parallel runs with high-current conductors.

RS-485 Cable

- **Construction:** twisted pair for **A/B**
- **Characteristic impedance: 120 Ω** (recommended)
- **Shielding:** overall shield is recommended in cabinets; use individually shielded pairs + overall shield in high-EMI
- **Examples: J-Y(ST)Y 2×2×0.5 mm²** or **LI2YCY PiMF 2×2×0.50**
- Use one twisted pair for A/B. Run COM to every node. Required, not optional: the RS-485 port is not galvanically isolated.

Shield Grounding

- **Default recommendation:** bond cable shield(s) to cabinet **PE/EMC ground at the PLC end only**.
- **Do not** connect shields to signal terminals (RS-485 A/B/COM).
- If both ends are in equipotential bonded cabinets, shields may be bonded at both ends using proper 360° clamps.

Compliance & Certifications

The ENM-223-R1 is CE marked and designed to comply with the applicable European Union directives. The manufacturer, ISYSTEMS AUTOMATION (HomeMaster brand), maintains the technical documentation and a signed EU Declaration of Conformity (DoC).

EU Directives

EMC 2014/30/EU · LVD 2014/35/EU · RoHS 2011/65/EU

Harmonised Standards

Area	Standards
EMC	EN 61000-6-1 (Immunity) · EN 61000-6-3 (Emissions)
Electrical Safety	EN 62368-1
RoHS	EN IEC 63000

Safety notice

L1 / L2 / L3 / N / PE: mains measurement inputs — hazardous voltage · 24 V DC: SELV · Qualified personnel only

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